

University of Eloued  
Academic Year: 2025/2026

Lesson: Linear Regression and Logistic Regression

Dr. Medini Atmane  
University of Eloued  
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Short Summary and Series

**Logistic regression**

**Party Two: Logistic regression Example**

**1.first: dataset**

Depended variable: Pass

| Observation | Hours_Studied (X) | Pass |
|-------------|-------------------|------|
| 1           | 1                 | 0    |
| 2           | 2                 | 1    |
| 3           | 3                 | 0    |
| 4           | 4                 | 1    |
| 5           | 5                 | 1    |

- **Example 1:** Simple Logistic Regression with one independent variable: Study\_Hours
- **Example 2:** Multiple Logistic Regression with five independent variables.

**Example 1 — Simple Logistic Regression( use Coding python method)**

Variables:

- Independent variable: Study\_Hours
- Dependent variable: Pass((0=Fail, 1=\Pass))

**Dataset:** you can find the dataset in [Git\\_Hub link](#)

**Questions**

1. What are the dependent and independent variables?
2. Write the logistic regression model.
3. Estimate the coefficients (beta\_0) and (beta\_1).
4. Write the estimated logistic regression equation.
5. Calculate the probability of passing for a given number of study hours.

6. Classify the observation as **Pass (1)** or **Fail (0)** using a probability threshold of 0.5.

*Note : Let the Coefficients take the next values:*

**const**      2.6486

**x1**          1.0904

7. Calculate the prediction errors/classification results.
8. Calculate the confusion matrix.
9. Calculate Accuracy.
10. Calculate Precision.
11. Calculate Recall.
12. Calculate F1-score.
13. Estimate the model in Python using `statsmodels` or `scikit-learn`.
14. Compare the manual and Python results.

## **Second Part:**

Remade the same work using Multiple Logistic regression